Project Completion Report

on

"Speed breaker and Pothole Detection and Alert System "

Submitted by

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Under the Guidance of

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At



DEPARTMENT OF ECE(AI&ML) ENGINEERING

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Certificate of Completion

This is to certify that Mr. Dhruv Shailendra Abhyankar *(name) with PRN number 1032220065, of class* ***TYECE(AI&ML)*** has successfully completed **Project Work** during the period of 09/08/2024 *(from date) to 20/04/2025 (to date).The report is being submitted towards partial fulfilment of Integrated BTech program in Electronics & Telecommunication Engineering for academic year* ***2024 - 25***.

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Project Report



Vision

To be an acclaimed institute which facilitates to earn enriching knowledge, skills and life-long learning in the streams of engineering to promote culture of world peace.

Mission

To facilitate our students and faculty with exposure and platform to contemporary practices to meet various challenges in ever advancing sectors of their field while using the gained knowledge for the welfare of human being.

To imbibe personal attributes and facilitate the students to pursue higher education and / or venture in to entrepreneurship avenues by building competence to comprehend society / social needs

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Abstract:

Road obstacles such as speed bumps and potholes are significant contributors to accidents in the World, where around 60,000 accidents were reported due to potholes between 2014 and 2022 alone. Advancing safety technologies have been implemented such as ADAS (Advanced Driver Assistance System). Although their effectiveness has been proven through real-life scenarios, ADAS has trouble detecting smaller objects such as potholes or speed breakers and they are mainly present in high -end luxury vehicles.

Measures are taken using machine learning models integrated with camera and ultrasonic sensor to reduce such accidents. A camera is used to provide the machine learning model with a visual representation of the world in front of it. The ultrasonic sensor is used to measure the distance from the obstacle. A Raspberry Pi 5 stores and processes data using YOLOv5s algorithm, which is trained using a labelled dataset consisting different types speed breakers and potholes. These components allow for a cheaper and very reliable system capable of accurate detection.

The two sensors work together to provide data to the microprocessor which feeds the camera output to YOLOv5s algorithm and uses ultrasonic sensor to detect elevation. A simple code will give an output based on the two given parameters and when a speed bump or pothole is detected, it initiates a buzzer to alert the user. The model car on which the system is mounted on uses an Arduino Uno R3 connected with a Bluetooth module. The system was tested by driving it down in a controlled as well as real world environment. The precision for detection is very high.

Chapter 1 : Introduction

1.1 Background

Potholes and unmarked speed breakers are two types of road hazards that cause significant risks to safety of vehicle drivers and passengers. Due to the absence of precise detection and alert systems, these hazards cause severe accidents and fatalities. This issue becomes worse in medium-to-budget segment vehicles because of their low safety ratings.

The objective of this project is to develop a system which affordable with the use of a camera, ultrasonic sensors, and a Raspberry Pi microprocessor to provide real-time alert when a speed breaker or pothole is detected. According to statistics, around 60,000 accidents were reported due to potholes in India between the years 2014 and 2022. Furthermore, over 30,000 accidents caused by speed breakers between the years 2014 and 2016, resulting in nearly 31,000+ fatalities during this period.

Due to the high implementation cost of these systems, they are currently equipped only in certain luxury vehicles, such as those from Mercedes-Benz and Range Rover. Our project aims to design a cost-effective system that will enable the manufacturers to include this essential safety feature into all vehicle segments, to enhance road safety for everyone.

1.2 Problem Statement

Drivers need a speed breaker and pothole detection and alarm system to enhance road safety by receiving real-time hazard alerts to take appropriate action to avoid accidents and fatalities.

1.3 Need/ Motivation

Every year, severe accidents occur due to unmarked speed breakers and potholes, especially under low-visibility conditions. In India, these hazards cause thousands of fatalities each year, along with vehicle suspension damage, traffic congestion, and other associated costs. Furthermore, driving over such hazards at high speeds can cause severe health problems like spinal compressions and lower back pains.

Our system would reduce these risks by providing alerts to drivers, which would help them to take appropriate action before driving over such hazards. Hence, this system will increase road safety and will lead to a better quality of life by reducing such accidents, health problems and vehicle repair costs.

1.4 Intended Audience

The intended audience of this project includes various users who can benefit from speed breaker and pothole detection and alert system. These include:

- **Private and Commercial Vehicle Owners:** To receive real-time alerts and avoid damage caused by going over potholes and speed breakers at higher speeds and enhancing road, vehicle safety and comfort.
- **Government and Road Maintenance Authorities:** To assist in timely road repair and maintenance through automated data collection and reporting.
- **Automakers:** To integrate this affordable system into budget segment vehicles as an additional road safety feature instead of expensive ADAS.

1.5 Project Scope

Overview

This project aims to design and develop a system to provide real-time alerts when potholes and speed breakers are detected to reduce accidents and fatalities. This system will use camera and an ultrasonic sensor to achieve it.

Objectives

The objectives of this project are to detect speed breakers and potholes with 90% accuracy and to provide real-time alerts to the driver of the vehicle.

Scope Inclusions

- This system will include the detection of speed breakers and potholes using sensor technology and an alert system with the use of a buzzer and to give visual alerts using 3 LEDs.
- The scope covers testing the system in urban areas to evaluate performance and reliability.

Scope Exclusions

- This project does not include GPS functionality.
- Data collection and storage is not included.

- Physical repair/ maintenance of potholes is not included.

Constraints

- The project will be completed within 6 months.
- The budget of the project is limited to ₹10,000 and the system will be tested in urban environments only.

Assumptions

- The system will be efficiently able to detect hazards from farther distances.
- The false positives of the system would be lesser with the inclusion of an ultrasonic sensor.

Chapter 2: Literature Survey

[1] This research presents a robotized Raspberry Pi Based System for Pothole 3D Reconstruction and Mapping. Potholes pose a serious risk to transportation safety and infrastructure. They lead to accidents, vehicle damage, and increased maintenance costs. Traditional road inspection methods are labour-intensive and were especially impractical during the COVID-19 pandemic. To address this, a deep learning-based system using YOLOv5 was developed for automated, real-time pothole detection and measurement. The model architecture includes CSPDarknet as the backbone for feature extraction, PANet for multi-scale feature aggregation, and a detection head those outputs bounding boxes and class probabilities. A custom dataset was created featuring potholes under various conditions. This includes water-logged and dry surfaces, diverse shapes and sizes. Images were annotated and used to train the model, which detects potholes in both images and video streams. It also estimates pothole dimensions using triangular similarity. Implementation involves training the YOLOv5 model with custom data and using the detect.py script for deployment. The proposed model achieved 90% accuracy with significantly reduced training time (15 minutes) and lower dataset size (27MB). This makes it an efficient and suitable model for real-time applications. Future improvements include detecting manholes, integrating GPS for location tagging, deploying the system in vehicles, and enhancing dataset quality.

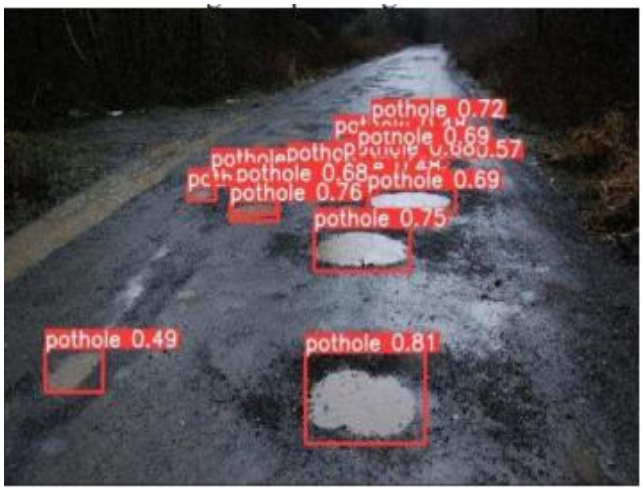


Fig 2.1 : Paper [1] Output of model

[2] This research introduces a portable, low-cost system for smart pothole detection and reporting model. It is specifically aimed at improving road safety in developing countries. Traditional detection systems often rely on smartphones or costly hardware, making them impractical for widespread, real-time use. To address this, the proposed solution utilizes a Raspberry Pi microcontroller paired with a camera module for image capture, a GPS module for precise location tagging, and LEDs to indicate detection and system status. The system is powered via the vehicle's lighter socket and requires no smartphone to function. Images of the road are continuously captured and processed in real time using image-processing algorithms based on OpenCV. The pipeline includes noise reduction, edge detection using the Canny method, and contour filtering to validate the size and shape of potential potholes. Upon detection, GPS coordinates are linked to the pothole image, and data is automatically uploaded through a portable Wi-Fi connection. First it is sent to Dropbox, then to a web server for centralized reporting. The server interface displays pothole images along with their GPS locations. In controlled (ideal) environments, the system achieved 100% accuracy across speeds from 10–40 km/h. Under typical driving conditions with traffic and environmental disturbances, the accuracy drops to 87.45% due to false positives from elements like water puddles and reflections. The device processes at eight frames per second and responds to pothole detection within an average of 0.1244 seconds. The design is cost-effective, portable, and delivers real-time detection with high accuracy, even without smartphones or complex infrastructure. Though the system performs well, reducing false positives in less ideal environments remains an area for future improvement. Overall, this solution proves to be a practical and efficient approach to pothole monitoring and road safety enhancement.

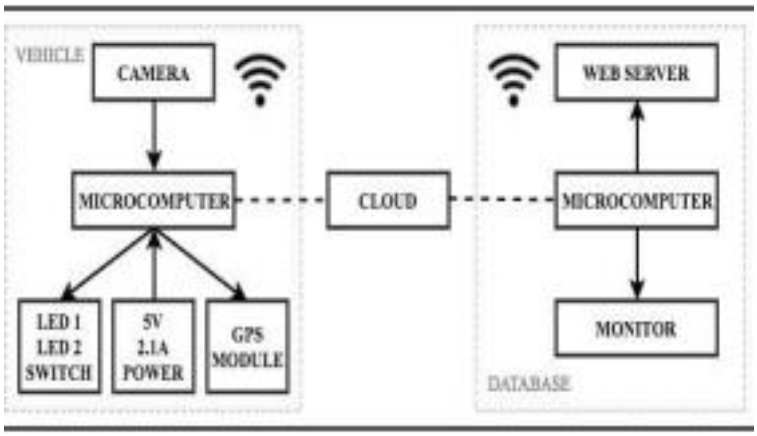


Fig 2.2 : Paper [2] Block diagram of system

[3] This research presents a low-cost IoT-based system designed for the real-time detection of potholes and speed humps on roads, with the objective of alerting drivers in advance to prevent accidents and improve road safety. Poor road maintenance contributes significantly to road fatalities, especially in countries like India where road transport accounts for the majority of passenger and freight movement. The proposed system offers a practical solution to this issue. It utilizes a PIC 16F877A microcontroller integrated with ultrasonic sensors to measure variations in road surface levels, distinguishing potholes and humps based on distance thresholds. A GPS module logs the exact location of these irregularities, and a GSM modem transmits the data to a server module. The server, hosted on an Android device, processes this information and stores it in a cloud database. A mobile application then uses this data to provide real-time alerts, both visual and audio beeps to drivers within a 100-meter range of the detected hazard. If the sensed distance exceeds a set threshold, the system identifies a pothole; if it is below the threshold, it is identified as a bump. This intelligent system not only enhances driver safety by providing timely alerts but also supplies valuable data to government authorities for road condition monitoring and maintenance. The system is cost-effective, quick to deploy, and leverages IoT and cloud technology for efficient communication and data handling. Future enhancements may include voice alerts and integration with more advanced navigation systems to improve usability and effectiveness. Overall, the solution offers a practical and scalable approach to mitigating road accidents caused by poor surface conditions.

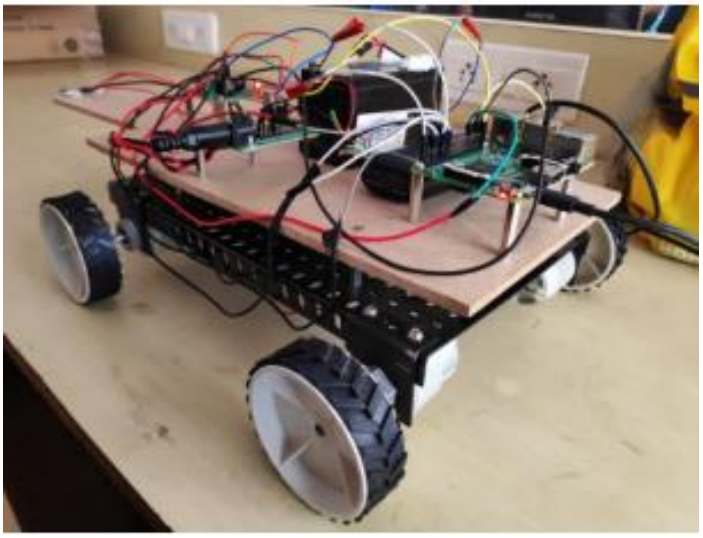


Fig 2.3 : Paper [3] Model image

[4] This research addresses the increasing problem of potholes on Indian roads. This is due to poor construction practices and inadequate drainage systems. These potholes pose a serious threat to drivers, especially at night or in low-visibility conditions. Given India's vast size and population, the frequency and impact of such accidents are significantly high. While the ideal solution would involve reconstructing roads with proper drainage infrastructure, this process is time-consuming and hindered by challenges due to our current system. As an alternative, this paper proposes a detection-based solution that avoids potholes altogether using real-time image processing. The system utilizes a camera module connected to a Raspberry Pi to capture videos of the roads. These videos are broken down into frames, which are then converted to grayscale to simplify processing. Noise is reduced using averaging, Gaussian and median blurring techniques. Morphological operations are applied to achieve accurate edge detection. The processed images are then analysed using Otsu's thresholding method, enabling precise identification and measurement of potholes. This image-based detection approach offers a cost-effective and real-time solution for road hazard identification. In terms of results, the system demonstrates high performance using deep learning models such as InceptionResNetV2 and DenseNet201, achieving an accuracy rate of 89.66%, outperforming other tested models. The integration of Raspberry Pi makes the system compact, affordable, and compatible with other modules. Additionally, the use of 2D visualization enhances the interpretability and practical application of the system. Overall, this solution enables efficient pothole detection and supports real-time hazard avoidance, contributing to safer road travel and potentially reducing accident rates also.

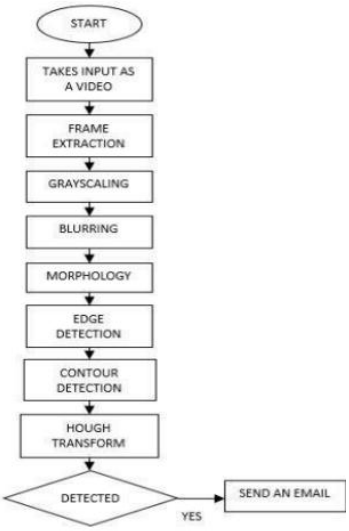


Fig 2.4 : Paper [4] Process flow of model

[5] This research introduces a low-cost, robotic system for 3D pothole mapping using a Raspberry Pi-based platform mounted on a remote-controlled robot. It is designed to capture geo-tagged images for constructing 3D models of potholes and calculating their volume for efficient repair planning. The system uses a Raspberry Pi 4 Model B with 8GB RAM, a GPS module, and a camera mounted on a chassis, which moves in a circular path around potholes to capture images at regular intervals. Python scripts automate the process of image capture and GPS tagging every 5 seconds, pausing the robot periodically to ensure image stability. These images are then processed to generate accurate 3D models of potholes, allowing volume estimation to determine the amount of fill material needed. The spatial data, including pothole location, volume, and accompanying images, are mapped using QGIS software for visual representation and planning. To evaluate the effectiveness of the system, both controlled lab simulations and real-world field tests were conducted. This includes trials in places such as a private parking lot, etc. This demonstrates the system's ability to function in uncontrolled outdoor conditions. The robot travels at a steady speed of 0.4 m/s. It efficiently maps potholes without human intervention, reducing the risk to maintenance workers. The system proved reliable in collecting spatial data and creating volume-accurate 3D pothole profiles. Conclusively, this approach provides a scalable, affordable, and safer alternative for road condition monitoring and maintenance, improving overall road safety while leveraging accessible hardware and open-source tools.



Fig 2.5 : Paper [5] Prototype Image

Chapter 3: Requirements

3.1 Functional Requirements

Object Detection

The system must be able to accurately detect presence of speed breaker and pothole using camera and the YOLOv5s model must classify objects in correct category.

Distance Measurement

The ultrasonic sensor must measure the distance towards the detected object.

Alert System

The system must trigger the buzzer when speed breaker or pothole is detected and when it is within 120 cm range.

Real-Time Processing

The system must process the video frames from camera and the sensor data in real-time

Visual Alert

The system must turn on the red LED upon start of service, must blink the yellow LED upon detection using camera and must turn on green and yellow LEDs upon the confirmation using ultrasonic sensor.

3.2 Non-Functional Requirements

Performance

The YOLOv5s model must achieve minimum 80% detection accuracy(mAP@0.5) and latency of detection must not exceed 0.5 seconds per frame.

Temperature

The temperature of Raspberry Pi 5 must remain under 65 degrees Celsius to maintain the response time of the system.

Energy Efficiency

The system must be able to operate on the available power of 27W power adapter.

Reliability

The system must work with constant accuracy and precision under day/night conditions and should be able to handle different road conditions such as wet, dusty, etc. with minimum false predictions.

Ease of Deployment

The system must be easy to deploy on Raspberry Pi and minimum external dependencies.

Cost-Effectiveness

The overall cost of system must remain within the budgets.

3.3: Hardware Requirements

Component	**Details**
Microprocessor	Raspberry Pi 5 8GB RAM
Camera Module	LG AN-VC500 Webcam 1080p
Buzzer	PCB Mounted Active Buzzer (1.5V - 15V operating voltage)
Ultrasonic Sensor	HCSR-04 Ultrasonic Sensor
Raspberry Pi Adapter	Raspberry Pi 5 Adapter 27W 5.1V 5A
Wiring	Jumper Wires, Copper Wires
Microcontroller	Arduino Uno R3 to drive the chassis.
Motors	Single Shaft Gear Motors 300 RPM
Wheels	Wheels for BO Motor
Motor Driver	L298N Motor Driver
Bluetooth Module	HC-05 Bluetooth Module
Chassis/Casing	Medium Density Fiber Board

3.4: Software Requirements

Software	Purpose
Raspberry Pi OS	Operating System for Raspberry Pi based on Debian Bookworm.
Arduino IDE	To code Arduino to drive the chassis.

3.5: Library Imports

Library	Environment
python3-pip	System-level
libopenblas-dev	System-level
libopenmpi-dev	System-level
libomp-dev	System-level
Ffmpeg	System-level
Numpy	Python (pip)
opencv-python	Python (pip)
Onnxruntime	Python (pip)
RPi.GPIO	Python (pip)
Gpiozero	Python (pip)
Arduino AVR Boards	Arduino IDE

Numpy (Python - pip):

Powerful library for numerical computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays.

opencv-python (Python - pip):

A library that allows you to work with computer vision tasks. It provides tools for image and video processing, such as detecting objects, reading images, and applying transformations or filters.

Onnxruntime (Python - pip):

A cross-platform, high-performance library for running machine learning models in the ONNX (Open Neural Network Exchange) format. It provides fast inference on various hardware accelerators like CPUs, GPUs, and specialized AI chips.

RPi.GPIO (Python - pip):

A library to control the GPIO (General Purpose Input/Output) pins on a Raspberry Pi. It enables interaction with physical components such as sensors, motors, and LEDs by allowing Python scripts to read from and write to the Pi's GPIO pins.

Gpiozero (Python - pip):

Another library for controlling Raspberry Pi GPIO pins. It simplifies the process of interacting with hardware like LEDs, buttons, and sensors through an object-oriented approach, making it more beginner-friendly than RPi.GPIO.

Arduino AVR Boards (Arduino IDE):

A package for the Arduino IDE that supports Arduino boards based on the AVR microcontroller family (such as Arduino Uno). It provides the necessary software to program and interface with these boards for embedded projects.

3.6: Feasibility Study

Technical Feasibility – This project requires uses Ultrasonic Sensor for distance measurement and camera for image recognition which are controlled by Raspberry Pi and the modules are compatible with each other hence this project is technically feasible.

Operational Feasibility – The YOLOv5s model gives a precision of 83% which is optimal for the project. Processing power of the microprocessor is enough to handle the model and provide real-time alerts.

Economic Feasibility – Since this system is a prototype, implementation costs are lesser. A practical model would require better hardware which will in turn increase the overall cost.

Social Feasibility – The system improves road safety by providing real-time alerts to the vehicle drivers regarding incoming potholes and speed breakers. This helps the driver to take appropriate actions which reduces accidents, fatalities, vehicle damages, etc. This would promote cautious driving and will lower the vehicle damages. Hence it would be socially feasible and widely acceptable.

Chapter 4: Action Plan

Sr. No.	Work Done	Schedule
1.	Purchasing of components and block diagram design.	Dec 2024 – Jan 2025
2.	Fine tuning of YOLOv5s Model over training dataset.	Jan 2025 – Feb 2025
3.	Coding and making of model car to drive the system.	Feb 2025 – March 2025
4.	Code implementation, prototype designing and stage 1 testing.	March 2025 – April 2025
5.	Improvements in system for better response time and accuracy and final testing.	April 2025 – May 2025

The project was executed in determined phases over a well-defined timeline. From December 2024 to January 2025, the focus was primarily on purchasing the required components and designing the block diagram of the system. In the next phase, January to February 2025, the fine-tuning of the YOLOv5s model was performed using a training dataset to optimize its accuracy for speed breaker and pothole detection.

From February to March 2025 coding was done and the model car was built that is used to drive the system. During the next phase from March to April 2025, the focus was primarily on implementing the code, making finishing touches to the prototype, and conducting Stage 1 testing to check and evaluate the system's functionality.

In the final phase, from April to May 2025, certain improvements were made such as changing to better heat sink and power supply to enhance the overall response of the system and even the accuracy based on feedback from earlier testing. This will be followed by final testing to ensure the system is fully operational and ready for deployment.

Gantt Chart

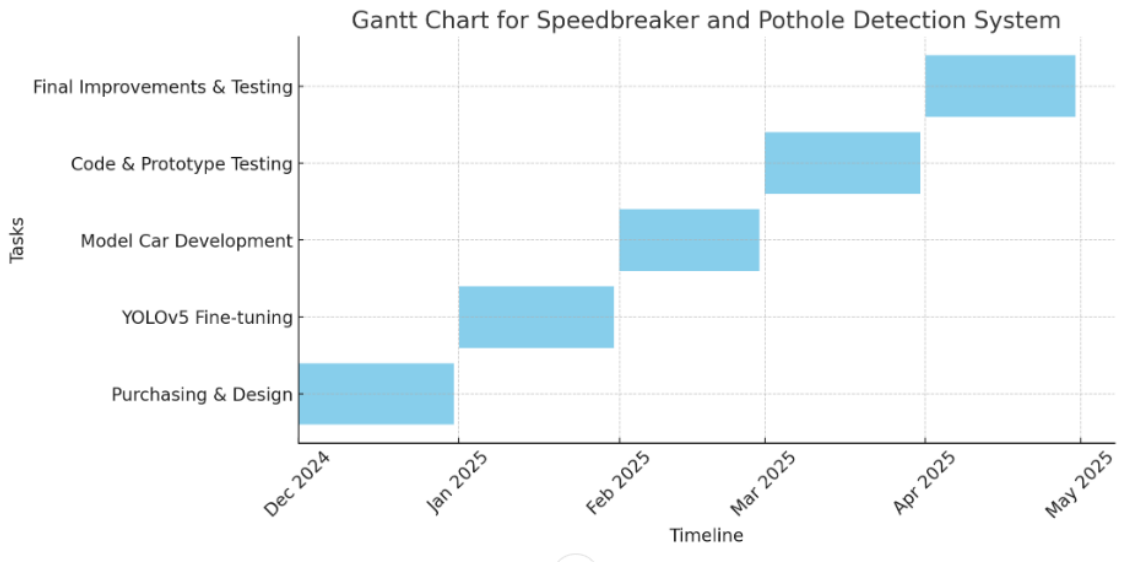


Fig 4.1 : Gantt Chart of System

Chapter 5: Proposed Works

5.1 Problem Definition

Unmarked speed breakers and potholes are certain road hazards that cause severe accidents and fatalities. These are even riskier under low visibility conditions for instance, at nights or in rainy or foggy weather. If a vehicle travels over such a hazard at high speeds, it can experience mishandling, tyre bursts, damage to vehicle suspension systems, etc.

If the problem is unaddressed, the number of such accidents and fatalities will continue to rise. This will lead to increased vehicle repair and maintenance cost, traffic congestion due to accidents, increased pollution levels, and will then indirectly negatively impact the overall quality of life.

5.2 Features

1. Real-Time alerts upon detection of speed breaker or pothole.

2. Affordable system than current existing solution.

3. Provides improved detection precision due to the integration of ultrasonic sensor along with camera.

4. Pothole detection in wet conditions.

5.3 Goal/ AIM

The objective of this project is to develop a cost-effective and highly accurate system that utilizes machine learning and distance measurement sensors to detect speed breakers and potholes on roads. The system aims to enhance road safety by providing timely, real-time alerts to drivers, thereby reducing the risk of accidents caused by sudden obstacles or poor road conditions.

This solution is especially beneficial in areas where road conditions are unpredictable, signage is poor, or speed breakers are unmarked. Ultimately, the goal is to assist drivers in making safer driving decisions, reducing vehicle damage, improving road awareness, and contributing to overall road safety.

Chapter 6: System Description

6.1: Block Diagram

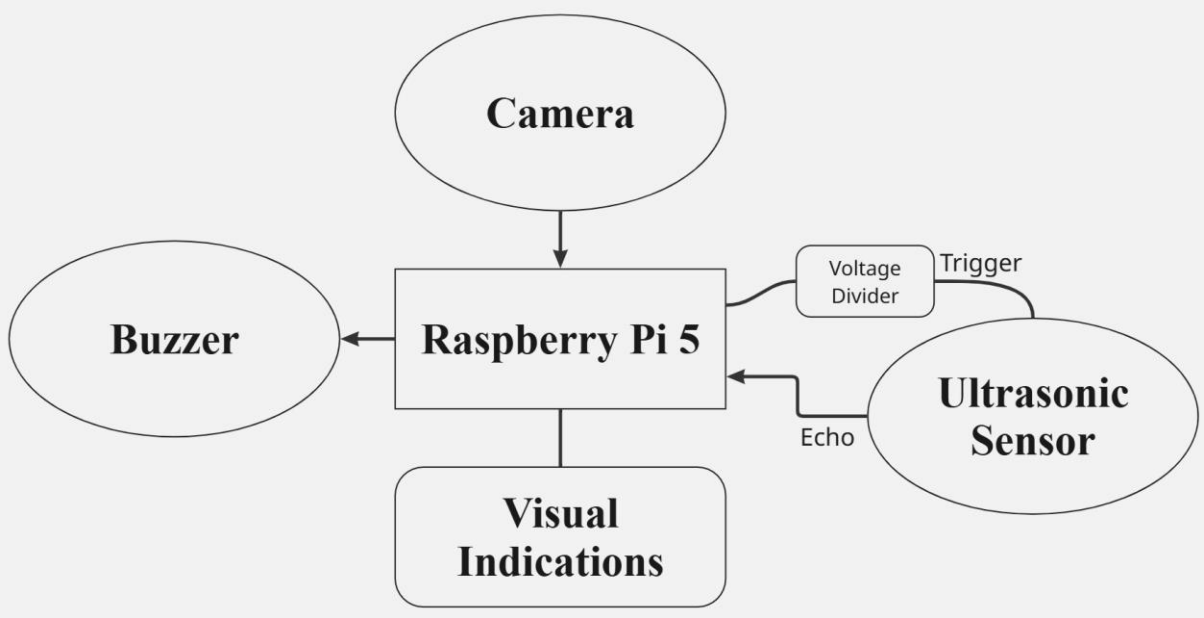


Fig 6.1 : Block Diagram of Proposed System

The block diagram consists of main microcomputer Raspberry Pi 5 8GB Ram in the centre. And certain peripherals such as the LG AN-VC500 Webcam, HC-SR04 Ultrasonic sensor, LEDs for visual alert. The LG AN-VC500 Webcam is connected to the micro computer using USB cord. The HCSR-04 Ultrasonic Sensor is connected to the microcomputer using jumper cables. The pin connections of Ultrasonic Sensor are as follows:

- Trigger Pin – GPIO 17
- Echo Pin – GPIO 27 (Along with voltage divider to keep voltage under 3.3V to limit the GPIO pin of Raspberry Pi from damaging)

The active buzzer is also connected directly using jumper cable to GPIO 25 of the Raspberry Pi. The LEDs for visual alert are connected as follows:

- Red LED – GPIO 5
- Yellow LED – GPIO 6
- Green LED – GPIO 26

6.2: Module Description

1. **Raspberry Pi 5**

The Raspberry Pi 5 is the latest and most powerful single-board computer from the Raspberry Pi Foundation, designed for high-performance computing in applications ranging from advanced personal projects and AI/ML development to IoT, automation, and industrial use. In this system, the Raspberry Pi 5 processes data from the camera and ultrasonic sensors using an integrated Machine Learning algorithm to control the buzzer based on object detection and distance measurements.

- Features a 2.4 GHz quad-core ARM Cortex-A76 CPU (64-bit), offering significantly improved performance over previous models.
- Equipped with 8 GB LPDDR4X RAM, enabling smooth multitasking and machine learning workloads.
- Supports Dual-Band Wi-Fi (802.11ac) and Bluetooth 5.0, along with Gigabit Ethernet for fast and reliable network connectivity.
- Includes two USB 3.0 and two USB 2.0 ports for connecting external peripherals and devices..
- Has a 40-pin GPIO header for flexible hardware interfacing and project customization.
- Offers Dual 4K Display support via two micro-HDMI ports, supporting up to 4Kp60 output.
- Includes hardware video acceleration for H.265 (4Kp60) and H.264 (1080p60) decoding.
- Powered via 5V/5A USB-C connector for stable and efficient power delivery.
- Comes with a microSD card slot for OS boot and data storage, and supports PCIe 2.0 expansion via FFC connector for high-speed peripherals.

The Raspberry Pi 5's enhanced processing power and expanded memory make it ideal for AI/ML applications that require fast data handling and real-time responsiveness.

2. **LG AN-VC500 USB 1080p Webcam**

   The LG AN-VC500 is a USB-based 2-megapixel Full HD webcam used for capturing real-time video input for object detection. It connects to the Raspberry Pi via USB and provides reliable, high-quality imaging for the ML algorithm to process.

   - Resolution: 1080p at 30 fps
   - Sensor: 2 MP
   - Interface: USB 2.0 (Plug and Play)
   - Wide Field of View for capturing road features
   - UVC Compatible – works with Raspberry Pi OS without additional drivers

   This camera is ideal when a USB-based imaging solution is preferred over CSI modules.

3. **HCSR-04 Ultrasonic Sensor**

   The HC-SR04 Ultrasonic Sensor is a widely-used and cost-effective distance measurement module known for its reliability and ease of integration. It operates using pulse-width modulation to accurately measure distances by emitting and receiving ultrasonic waves. The sensor will be utilized to calculate the distance between the module and nearby obstacles, which can then be fed into a Machine Learning algorithm to enhance object detection accuracy and responsiveness.

   - Distance Range: 2 cm – 400 cm
   - Accuracy: ± 3 mm
   - Operating Voltage: 5V DC
   - Operating Current: ~15 mA
   - Measurement Angle: ≤ 15°
   - High Sensitivity and Stable Performance
   - Simple 4-Pin Interface
     - VCC – Power Supply (5V)
     - GND – Ground
     - Trig – Trigger Input

- Echo – Echo Output

The HC-SR04 is best suited for applications where consistent obstacle detection is required and is ideal for robotics, automation, and embedded systems. It's simple interfacing and reliable range detection make it a valuable component for distance sensing projects.

4. **PCB Mounted Active Buzzer**

   It is compact sound actuator which will make a loud sound depending on the voltage input. It has an Input Voltage range of 1.5V to 15V. It is used as the alert indicator when the Algorithm detects a Speed breaker or a Pothole.

5. **LEDs for Visual Alert**

   Three Light Emitting Diodes (LEDs) of 5mm in colours red, yellow and green are used to provide visual alerts to the driver. The red LED provides alert when the script starts running, the yellow LED blinks when the camera detects speed breaker or pothole and both yellow and green LEDs turn on when the ultrasonic sensors confirms that a speed breaker or a pothole is there.

6.3: Advantages

- **Improved Road Safety**

  The system can alert the drivers in case of a presence of a speed breaker or a pothole in real-time, which will help in reducing accidents and fatalities.

- **Cost-Effectiveness**

  It uses affordable components and is cheaper than commercial grade solutions.

- **Portability**

  The components used in this system are portable.

- **Alerts**

  The buzzer creates a loud distracting sound and the visual alert LEDs provide visual indications as well which guarantees the drivers attention to both the sound and the visual indication.

- **Flexibility**

  The combination of the camera with the machine vision allows detection of obstacles in many conditions such as rain, etc.

6.4: Disadvantages

- **Detection range**

  Since the HCSR-04 sensor has a range of only 4.5 meters. At high speeds this will not be able to detect obstacles fast enough
- **Environmental factors**

  Conditions such as dust or extreme heat will reduce the reliability of the system.
- **Maintenance**

  The components will require frequent cleaning and calibration to prevent damage or a drop in performance.
- **Physical integrity**

  The physical integrity of the components may not be enough as the vehicles sudden shocks and vibration may impact the output readings of the sensors.
- **Complexity**

  It is difficult to integrate multiple components into a small system.

Chapter 7: Software Design

7.1 Introduction

The software design of Speed Breaker and Pothole Detection and Alarm System includes design, process flow and overall interaction between hardware and software. The system uses computer vision and sensor data to detect road hazards and provides real-time alerts to the vehicle driver.

7.2 YOLOv5s Design

The software uses a YOLOv5s model. The model was trained on a Windows PC. It was fine-tuned over a 769 MB dataset from Kaggle of 1400+ images of speed breakers and potholes. The images were divided into an 80/20 split. 80% was used for training and 20% was used as validation dataset. The model was tuned for 50 epochs and took around 15 minutes to train one epoch. The model was then exported to Onnx and the exported best.onnx was utilized in software. Then a python script was used in Raspberry Pi OS 64bit based on Debian Bookworm to process image frames from YOLOv5s, comparing it with Ultrasonic Sensor output and to alert using buzzer based on the prediction output.

7.3 Process Flow

The process flow is as follows:

- Power is provided to Raspberry Pi
- The Raspberry Pi, HCSR-04 Ultrasonic Sensor, USB Webcam, Buzzer and visual alert LEDs are initialized.
- The Red LED turns on indicating that the script has started executing.
- The camera starts capturing frames.
- The frames are processed using best.onnx of YOLOv5s.
- If the model does not detect speed breaker or pothole, the camera keeps on capturing frames.
- If the model detects speed breaker or pothole, the yellow LED starts blinking and the ultrasonic sensor HCSR-04 is used to measure the distance between the hazard and the model.
- If the distance is under 50 cm, the ultrasonic sensor analyses height if speed breaker is detected or depth if pothole is detected.

- Upon detection by ultrasonic sensor, a real time alert is provided by buzzer and both the green and the yellow LED turn ON.
- If ultrasonic sensor does not detect the hazard, the camera again starts capturing frames.

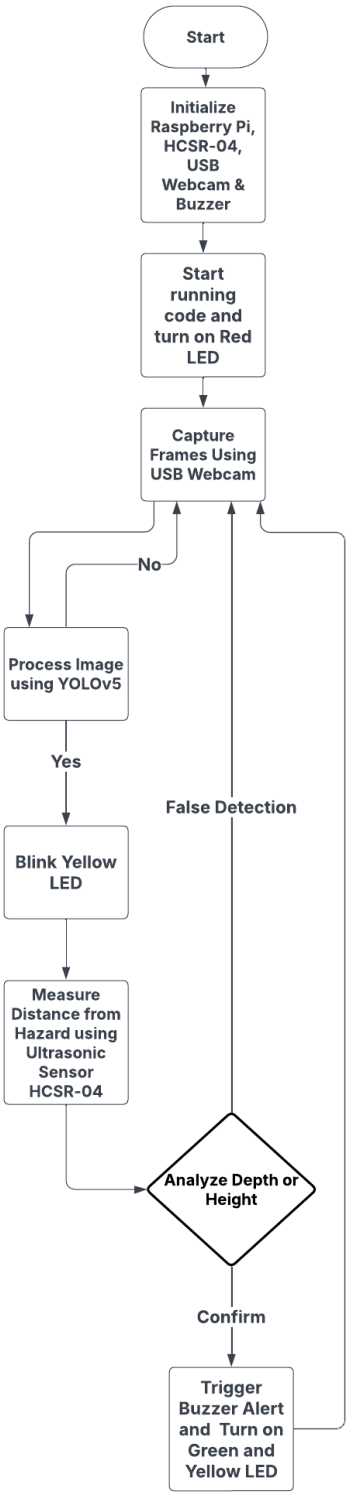


Fig 7.1 : Process Flow of Proposed System

7.4 Interface Design

There is no graphical display interface in this project. The system runs headlessly, but optionally logs detections or shows console output for debugging. But there is a visual interface using LEDs given for the driver that includes three LEDs to provide visual alerts to the driver.

Chapter 8: Coding and Testing

This system integrates computer vision with sensor-based validation to detect potholes and speed breakers using a Raspberry Pi 5. The coding phase involved model loading, camera integration, GPIO control, and real-time detection logic and the testing ensured reliable performance.

8.1 Key Components and Technologies Used

Coding Language Used: Python

Libraries Used: OpenCV, NumPy, ONNXRuntime and gpiozero

Hardware: Raspberry Pi 5, LG AN-VC500 Webcam, Ultrasonic Sensor (HC-SR04), Buzzer and LEDs

8.2 Code Functionality

Model Loading:

Loads a YOLOv5s ONNX Model for hazard detection.

Preprocessing:

Each frame from camera is resized, normalized and reshaped before it is passed to the model for detection.

Detection:

The model detects the objects as labels i.e. as pothole or speed-breaker with a confidence threshold of 50%. And the yellow LED starts blinking upon detection of hazard by the model.

Sensor Validation:

Upon detection by the camera, distance is measured using an ultrasonic sensor for physical confirmation of the hazard.

Alert System:

If the hazard is within a range of 120 cm from the vehicle, a buzzer is triggered to provide a sound alert to driver and visual alert is given by turning ON green and yellow LED.

8.3 Testing

The testing was performed in following ways,

- The test of only the detection system was performed using different printed images and certain real speed breakers and potholes to test the detection using camera.
- For validation using sensor, tests were performed to check sensor accuracy, whether sensor is able to differentiate between height and depth according to speed breaker and pothole and checked whether the distance measurement is accurate.
- Combined checks were performed by using final prototype on real speed breakers. The checks could not be performed for potholes due to limitations of power supply potability.
- Buzzer and LEDs successfully triggered within the specified distance range.
- Ensured smooth model inference using ONNX on CPU without lag.

8.4 Sample Console Output



Fig 8.1 : Console Output of System

8.5 Conclusion

According to the tests performed, the system runs reliably on Raspberry Pi 5, it handles real-time detection and physical alerting properly. GPIO control and sensor response were tested thoroughly and the desired functionality was successfully achieved.

Chapter 9: Results

9.1 Model Performance Metrics

After training the YOLOv5s model for 50 epochs, the best results were observed at Epoch 50, as shown in the bar chart below. The key performance metrics are:

- Precision: 83.69%
- Recall: 74.09%
- mAP@0.5: 74.09%
- mAP@0.5:0.95: 38.67%

These values indicate that the model is able to detect potholes and speed breakers with high accuracy and reliability. The high precision shows that the detections made are mostly correct, while the decent recall ensures that most actual objects are being detected. The mean average precision (mAP) values further validate the model's consistency across different confidence thresholds.

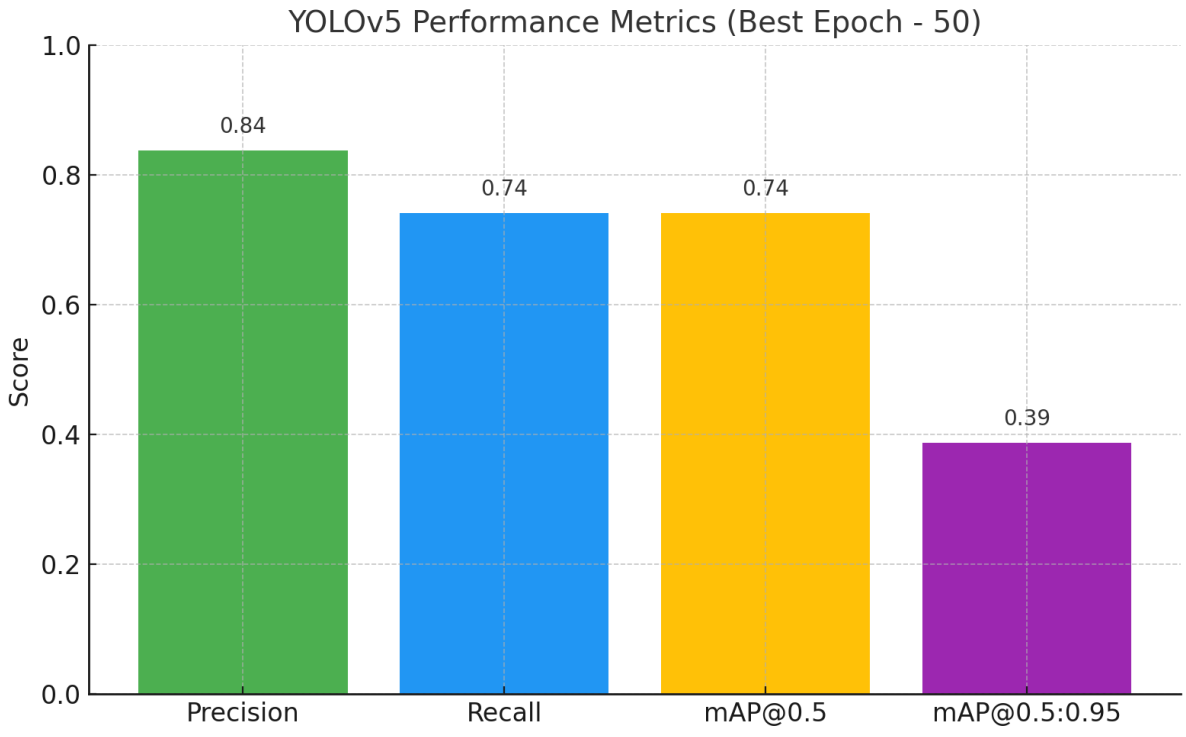


Fig 9.1 : YOLOv5s Performance Matrix

YOLOv5s Detection Images:

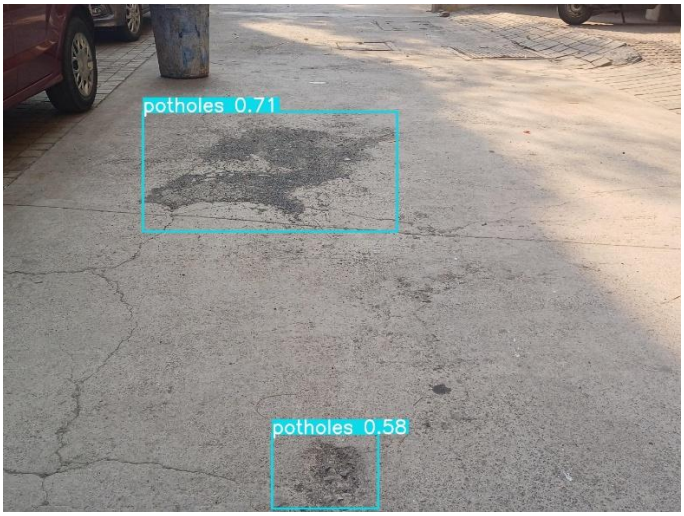


Fig 9.2 : The image shows confidence percentage of pothole detection

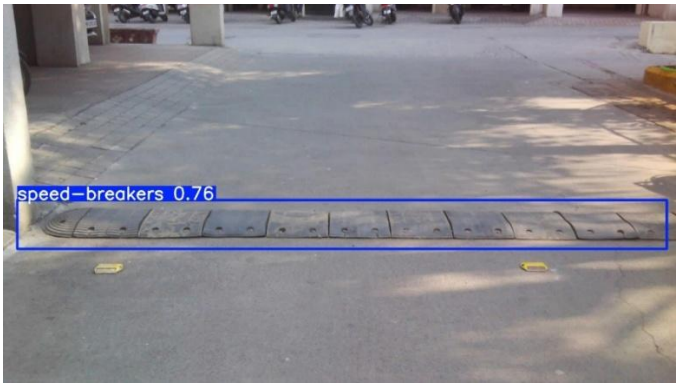


Fig 9.3 : The image shows confidence percentage of speed breaker detection



Fig 9.4 : The image shows confidence percentage of speed breaker detection

9.2 Response Time

The testing of the system was performed in an ideal environment due to the limitations of power supply in real-world conditions. These time responses are based on the time taken for the system to detect the speed breaker or pothole after it was in range of camera, confirm using ultrasonic sensor and triggering of buzzer from a distance of 120cm from the hazard. The testing was done 3 times each for speed breaker and pothole and the response time was recorded.

The response time when the system was stationary was as follows:

Reading No.	Object Distance from System	Response Time for Speed Breaker Detection	Response Time for Pothole Detection
1	120 cm	3 seconds	2.12 seconds
2	120 cm	2.3 seconds	2.57 seconds
3	120 cm	2.82 seconds	2.73 seconds

The tests were also conducted when the system was moving. The system was moving at around a speed of 4km/h. The system was placed at a distance of 130 cm from the prop of Speed Breaker and the detection occurred when the vehicle was 120 cm away from the Speed Breaker and the vehicle was stopped at a distance of 15 cm away from the Speed Breaker. The buzzer was triggered within 1.38 seconds since the ultrasonic sensor confirmed the presence of the Speed Breaker.

9.3 Visual Indications

The model consists of 3 LEDs for visual indication. The LEDs turn on as follows:

The Red LED turns on when the script starts running

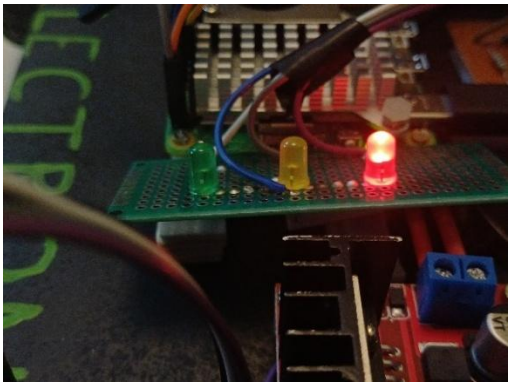


Fig 9.5 : Red LED turns ON when script starts running

The Yellow LED blinks when the camera detects the speed breaker or pothole

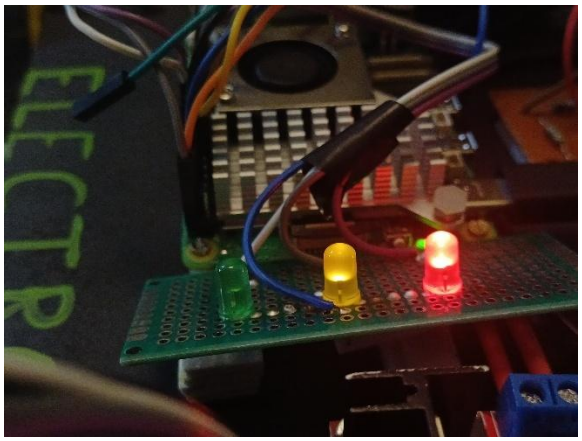


Fig 9.6 : Yellow LED blinks when system detects a speed breaker/ pothole

The Green LED turns on when the Ultrasonic Sensor confirms the presence of speed breaker and pothole.

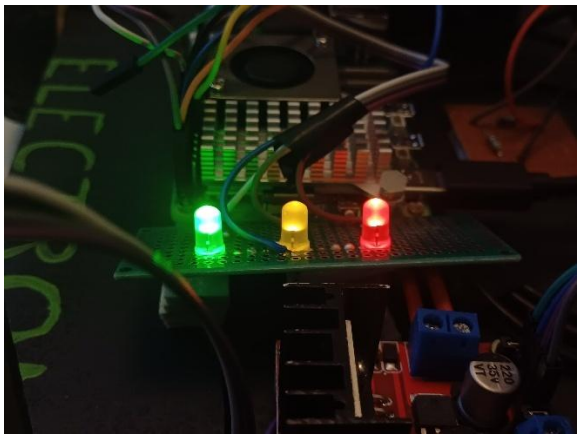


Fig 9.7 : Green and Yellow LEDs turn ON when ultrasonic sensor confirms detection

9.4 Prototype

Following figure shows the design of the prototype used for deriving above results:

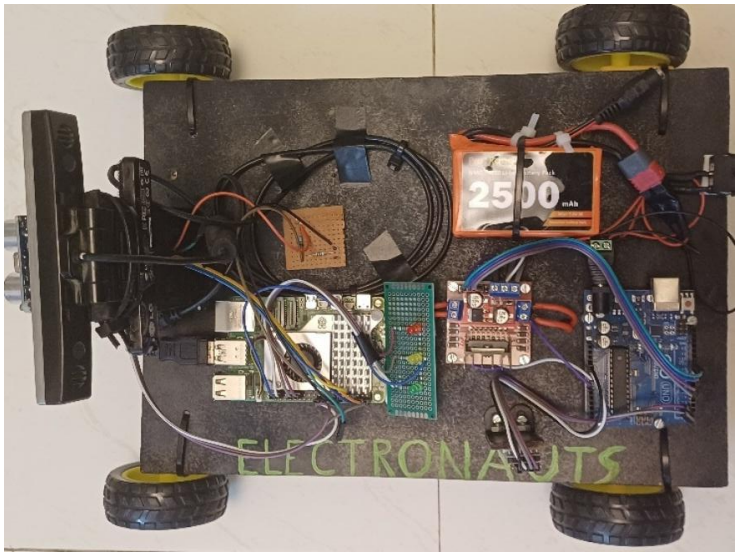


Fig 9.8 : Prototype Image

Conclusion:

The Speed breaker and Pothole Detection System developed in this project demonstrates a practical and innovative approach to improving road safety. It uses a combination of machine learning and sensor-based distance measurement using Camera and Ultrasonic Sensor. By accurately identifying road anomalies such as unmarked speed breakers and potholes in real time, the system effectively alerts drivers, helping them take quick actions to avoid potential accidents and vehicle damage.

This solution offers a low-cost yet efficient alternative to existing high-end systems, making it accessible for widespread adoption, especially in developing regions where road infrastructure is often unpredictable. The integration of technologies like camera-based obstacle detection, ultrasonic sensing, and real-time alert systems showcases the potential of combining AI and embedded systems for smart mobility solutions.

Overall, the project successfully meets its objective of enhancing driver awareness and promoting safer driving practices. It also lays the groundwork for future enhancements such as integration with navigation systems, smart cities, and autonomous vehicles, making it a scalable and future-ready solution for intelligent transportation systems

Future Scope:

The proposed model has the potential to evolve significantly beyond its current implementation. As road infrastructure continues to pose challenges, especially in developing countries such as India, the need for intelligent road condition monitoring systems will only grow. This system, which focuses on real-time detection and alerting can be enhanced and scaled for broader applications.

In the future, integration with GPS and mapping services can allow for the creation of dynamic road condition maps. These maps can be shared with other vehicles and centralized traffic management systems to improve route planning and traffic flow. Cloud connectivity will enable real-time data upload and identifying high-risk zones quickly. This will allow city planners and authorities to monitor road conditions to plan timely maintenance and repairs.

This system can be adapted for autonomous and semi-autonomous vehicles, acting as an additional layer of road awareness. By combining the detection data with vehicle control systems, future iterations can actively reduce speed or adjust suspension systems in response to detected potholes or speed breakers, improving both safety and comfort.

In terms of hardware, future versions can employ advanced sensors like LiDAR or stereo cameras to increase detection accuracy in various lighting and weather conditions. Moreover, by implementing edge computing, the system can process data directly on-device, reducing latency and dependence on network availability.

The project also opens up possibilities for crowdsourced road condition monitoring, where multiple vehicles contribute data, creating a robust, real-time database of road health. This will be especially useful for smart city initiatives and intelligent transportation systems (ITS).

Such systems may also be used in goods and transport vehicles which help drivers better focus on roads as long drives cause a lot of exhaustion and this causes accidents many times where there might not be any injuries but there will be a huge loss of goods due to damage. Amount of maintenance needed by trucks will also be reduced, decreasing costs needed to manage the trucks.

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